Hallucination detection for large language models at response, sentence, span and claim level: a structured review of signals, evaluation practice and evidence gaps (2022-2026)

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Abstract

Background: Large language models (LLMs) write false or unsupported statements in fluent text, and many automatic detectors have been proposed, each evaluated in its own way.

Objective: To map which signals hallucination detectors use, at which granularity they score output, and how well the published evidence supports comparison between them.

Methods: OpenAlex (four queries) and the arXiv API (two queries) were searched on 24 October 2026 and 33 records were added from other sources. Of 1,788 records identified, 746 were screened, 56 full texts were assessed and 54 primary studies (2022-2026) were included. Each study was coded in a 31-field extraction matrix and an eight-item quality checklist by a single reviewer.

Results: Of 46 detectors, attention-based (13), sampled-consistency (12) and hidden-state (10) methods were the largest groups. 22 of 46 scored whole responses and 9 scored individual claims. Only 1 claim-level detector was both training-free and single-pass, and it needs an auxiliary NLI model. Code was linked by 41 of 54 (75.9%) studies and own data released by 17 of 54 (31.5%). Variance or significance tests were reported by 15 of 54 (27.8%), transfer to new data, models or languages was tested by 16 of 54 (29.6%), compute cost was measured by 10 of 54 (18.5%) and evaluation beyond English appeared in 6 of 54 (11.1%). Only 7 studies used a protocol that allows direct comparison with other studies.

Conclusion: Detection methods are numerous but their evidence is hard to compare, so no ranking across papers is given. Shared claim-level benchmarks, cost reported against generation time, interval estimates and multilingual test sets are the main needs. Nine gaps are listed, each with counts and a proposed study.

Keywords: hallucination detection; large language models; uncertainty quantification; claim-level verification; attention; evaluation methodology; structured review

1. Introduction

LLMs are now used to answer questions, summarise documents and draft long reports. They also produce statements that read well but are false, or that are not supported by the source they were given. These outputs are usually called hallucinations. Because a hallucinated sentence looks the same as a correct one, a reader cannot catch it by style alone, and automatic detection is needed before LLM output is trusted in search, medicine, law or education.

Work on detection has grown quickly. In the included set the number of studies rises from 1 in 2022 and 5 in 2023 to 17 in 2024 and 20 in 2025, with 11 already in 2026 (Figure 2). The methods differ in nearly

every design choice. Some sample the model several times and measure agreement [6, 7, 54]. Some read attention maps or hidden states from the single forward pass that produced the text [3, 15, 16]. Others call an external verifier or a stronger LLM [22, 30, 41]. They also score different units: a whole answer, a sentence, a token span or an atomic claim. Since each paper chooses its own datasets, labels and metrics, the reported numbers are hard to put side by side.

Existing surveys (Section 2) cover hallucination in general, uncertainty quantification in general, or abstention. None of the five examined here codes each study for score granularity, compute cost, transfer testing and reproducibility, or counts how often these are reported. This review does that. It is a structured review with a documented database search. It is not presented as a full systematic review, because one reviewer screened the records and not every eligible full text was retrieved (Sections 4 and 10).

The review answers five research questions:

- RQ1. Which signal families are used to detect hallucinations, and how do they divide by supervision, granularity and cost?
- RQ2. Which datasets, labelling procedures and metrics are used, and how comparable are the reported results?
- RQ3. How often do studies measure the compute cost of detection, and which cost classes exist?
- RQ4. How often is generalisation tested across datasets, models and languages?
- RQ5. How reproducible is the evidence, judged by code release, data release and statistical reporting?

The contributions are: (i) a documented search and screening ledger whose counts can be traced to a workbook; (ii) a four-axis taxonomy of 46 detectors with an assignment rule for every category; (iii) a comparison table of all 54 studies without cross-setting ranking; (iv) counts of evaluation and reproducibility practice; and (v) nine evidence-based gaps with proposed studies. Section 2 compares existing reviews, Section 3 defines terms, Section 4 gives the method, Section 5 the taxonomy, Section 6 the comparative analysis, Section 7 datasets, metrics and evaluation, Section 8 challenges and limitations of the methods, Section 9 gaps and future directions, Section 10 the limitations of this review and Section 11 the answers to the research questions.

2. Existing reviews and how this review differs

Five recent reviews [55-59] were compared on scope, period, taxonomy, evaluation analysis and what they leave out (Table 1). Huang et al. [57] give the broadest account of hallucination, covering causes, detection, benchmarks and mitigation, with detection as one section among many. Shorinwa et al. [58] and Geng et al. [59] sort uncertainty and confidence methods into white-box and black-box classes. Geng et al. appeared before most attention-based and claim-level detectors, which in this set date from 2024 onward. Wen et al. [55] cover abstention, where detection is only one input to a refusal decision. Kang et al. [56] are closest to this topic and run their own experiments on two models, but they place attention methods inside a general internal-state group and leave claim-level detection to future work.

None of the five states its search strategy or search period. None counts how often primary studies report variance, compute cost, transfer or non-English evaluation. This review adds those counts, a granularity axis, a cost axis and a comparability code for each study.

Table 1. Comparison of existing reviews with this review.

Review	Year	Scope and period	Taxonomy	Evaluation analysis	Missing / what this review adds
R1 Wen et al., Know Your Limits (abstention survey), TACL 2025 (arXiv 2407.18418)	2025 (TACL; venue from OpenAlex record R0325)	Abstention from 3 views (query, model confidence, human values); lifecycle stages; general-purpose chatbots. Period/search method: NOT STATED	Yes (lifecycle x 3 perspectives)	Surveys benchmarks and metrics; no experiments of its own	Safety refusal dominates; detectors only as probing/uncertainty estimation; no claim/span granularity, no attention family, no comparability or reproducibility audit
R2 Kang et al., UQ for hallucination detection, arXiv 2510.12040	2025 (arXiv v1, 14 Oct 2025); preprint	UQ for hallucination detection on open-ended QA (short and long form); excludes agents and mitigation. Period/search method: NOT STATED; method table has 35 methods, newest 2025	Yes (4 axes: concept, sampling, accessibility, training reliance)	Yes: own experiments, TriviaQA/GSM8K (1,000 each) and FactScore-Bio; LLaMA-3-8B, GPT-4o-mini; AUROC, PRR. Long-form drops; verbalised confidence best	Covers attention methods but inside internal-state group; claim-level only as future work; 2 models; no reproducibility/variance/comparability audit
R3 Huang et al., A Survey on Hallucination in LLMs, ACM TOIS 43(2), DOI 10.1145/3703155	2025 (arXiv 2311.05232, 2023)	Hallucination in general LLMs: definitions, causes (data, training, inference), detection, benchmarks, mitigation, RAG limits. Period/search method: NOT STATED	Yes (factuality vs faithfulness; causes x stages)	Benchmark tables (size, language, label, metric); no experiments	Detection is one section; uncertainty/internal-state methods grouped coarsely; no granularity axis, no cost, no reproducibility or comparability audit
R4 Shorinwa et al., Survey on UQ of LLMs, ACM CSUR, DOI 10.1145/3744238	2025 (arXiv 2412.05563, 2024)	UQ methods for LLMs incl. token-level, self-verbalised, semantic-similarity, mechanistic; applications incl. robotics. Period/search method: NOT STATED	Yes (4 method classes)	Discusses benchmarks and metrics; no experiments	Claim-level detection and RAG faithfulness covered briefly; no counts of variance/code/data; no cross-paper comparability
R5 Geng et al., Survey of Confidence Estimation and Calibration in LLMs, NAACL 2024, DOI 10.18653/v1/2024.naacl-long.366	2024 (arXiv 2311.08298, 2023)	Confidence estimation and calibration for generation and classification; white-box vs black-box. Period/search method: NOT STATED	Yes (white-box logit/internal vs black-box consistency/verbal)	No experiments (authors list this as a limitation)	Pre-dates most attention-based and claim-level work (2024-2026); no hallucination-span granularity; no reproducibility audit

3. Background and terminology

The terms below are used in the same sense throughout.

- **Hallucination.** Generated content that is false with respect to world knowledge (a factuality error) or not supported by a given source such as a retrieved passage (a faithfulness error) [57].
- **Detector.** A method that assigns a score to a unit of LLM output so that hallucinated units score higher (or lower) than correct ones. Scores are compared with labels from humans, from an LLM judge, or from string matching against a reference answer.
- **Granularity.** The unit that receives a score: a whole response, a sentence, a token span or named entity, or an atomic claim (a short statement with one fact) [22].
- **Signal source.** What the detector reads. Output probabilities are the token probabilities assigned during generation; claim-conditioned variants such as CCP [2] adjust them for alternatives with the same meaning. Sampled consistency draws several answers and measures agreement, often in

meaning space, as in semantic entropy [6, 54]. Hidden-state probes train a classifier on intermediate activations [13, 16]. Attention-based methods use attention weights, for example the share of attention on the source context [15] or on sink tokens [5]. External verifiers split an answer into claims and check each against a reference with another model [22, 41]. Self-report asks the model for its own confidence [8].

- **White-box and black-box.** A white-box detector needs model internals (weights, activations or token probabilities); a black-box detector needs only generated text.
- **Training-free and supervised.** A supervised detector is fitted on labelled hallucinations; a training-free one uses a fixed formula or an existing model without new labels.
- **Metrics.** AUROC is the probability that a random hallucinated unit scores above a random correct one. PR-AUC (area under the precision-recall curve) is more informative when hallucinations are rare. The prediction-rejection ratio (PRR) measures how much output quality improves when the highest-scoring outputs are rejected [28]. Calibration error (ECE) compares stated confidence with observed accuracy.
- **Transfer.** Testing a detector on data, models, tasks or languages different from those used to fit, tune or calibrate it.

4. Review methodology

4.1 Design

The review follows the reporting items of PRISMA 2020 [60] where they apply to a single-reviewer course review. All decisions and counts are recorded in a workbook (Search_Log, Screening_Sheet, PRISMA_Counts, Extraction_Matrix, Quality_Check, Evidence_Summary, Gap_Table), which is supplied as supplementary material together with the exports of the verification re-run described in Section 4.2.

4.2 Sources and search

OpenAlex [61], an open index that covers ACL Anthology, IEEE, ACM, Springer and arXiv records, and the arXiv API were searched on 24 October 2026. Scopus and Web of Science were not available to the author through an automatable interface, which is listed as a limitation. OpenAlex queries searched title and abstract from 1 January 2020 to the search date. Table 2 gives the exact strings and record counts. For each OpenAlex query the 200 most relevant records were exported; 739 lower-ranked records were not exported. Thirty-three records came from other sources: 29 studies collected before the search, one found by citation chasing (Farquhar et al. [54]), two surveys and one off-topic paper from the prior collection.

All six queries were run again on 26 October 2026 to check that the search can be repeated. OpenAlex returned 391, 337, 409 and 420 records for OA1-OA4 (two days earlier: 389, 336, 401 and 413; the index grows daily), and 197, 199, 195 and 195 of the 200 originally exported records appeared again in the top 200. The arXiv queries returned the same 132 and 84 records. The re-run exports are kept with the workbook.

Table 2. Search strings as executed and records returned (24 October 2026).

ID	Source	Query	Records
OA1	OpenAlex	("hallucination detection" OR "uncertainty quantification" OR "confidence estimation") AND ("large language model" OR LLM) AND (claim OR "fine-	389

ID	Source	Query	Records
		grained" OR "long-form" OR "token-level" OR "atomic fact")	
OA2	OpenAlex	(attention OR "attention head" OR "hidden state" OR probe) AND hallucination AND ("large language model" OR LLM) AND detection	336
OA3	OpenAlex	(abstention OR "selective generation" OR "selective prediction" OR conformal) AND ("large language model" OR LLM) AND (hallucination OR factuality)	401
OA4	OpenAlex	(benchmark OR dataset) AND hallucination AND (claim OR span OR "retrieval-augmented") AND detection AND ("large language model" OR LLM)	413
AX1	arXiv	abs:"hallucination detection" AND (abs:claim OR abs:"fine-grained" OR abs:"long-form" OR abs:"token-level")	132
AX2	arXiv	abs:hallucination AND (abs:"attention heads" OR abs:"attention maps" OR abs:"hidden states") AND abs:detection	84
OS1	Other sources	No query: 29 primary studies already collected, Farquhar et al. 2024 (Nature) from citation chasing, 2 surveys and 1 off-topic paper	33

4.3 Eligibility criteria and quality questions

Table 3 lists the inclusion and exclusion criteria applied at screening and the eight quality questions applied to every included study, with the rule used to answer each.

Table 3. Eligibility criteria and quality-appraisal questions.

Code	Criterion or question	How it was applied
I1	Proposes, evaluates or benchmarks a method that detects, scores or abstains on hallucinated or incorrect content in LLM-generated text	Title, abstract, then full text
I2	Reports a quantitative evaluation	Full text
I3	Published from 2020 onward, in English	Search filter and record date
I4	Peer-reviewed paper or arXiv preprint	Venue field; preprints flagged
E1	Multimodal, speech or diffusion-model output	Title and abstract
E2	Secondary study (review, survey, editorial)	Title and abstract; used in Section 2 instead
E3	Non-archival grey literature (Zenodo, SSRN, Preprints.org, Research Square)	Source field
E4	Not about detection or uncertainty (mitigation, application or theory only)	Title and abstract
Q1	Is the objective clear?	Yes if the task and claim are stated
Q2	Is the method described well enough to re-implement?	Yes if inputs, features and training are given
Q3	Are the datasets described?	Yes if name, size, split and label source are given
Q4	Are baselines fair?	Partial if strong recent baselines are missing or tuned differently
Q5	Is the evaluation valid (no leakage)?	Partial if labels come from the same model family or test data overlap training
Q6	Are variance or statistics reported?	Yes for SD, standard error, intervals or significance tests
Q7	Are code or data available?	Yes if a working public link is given
Q8	Are limitations stated?	Yes for a limitations section or explicit paragraph

4.4 Screening and selection

After merging the exports and removing 303 duplicates by normalised title, 746 records were screened on title and abstract (Figure 1). 319 were excluded (E1 73, E2 32, E3 94, E4 120). Of 427 records eligible for full-text reading, 56 were retrieved and assessed. The other 371 were not retrieved because one reviewer could not read them all. The retrieved set was chosen by relevance rank, peer-review status and coverage

of method families that were thin in the prior collection. Two full texts were excluded (one off-topic, one secondary study used in Section 2), leaving 54 included primary studies. Title and abstract decisions were proposed with keyword rules for E1-E4 and checked by hand. The rules and every decision are in the Screening_Sheet.

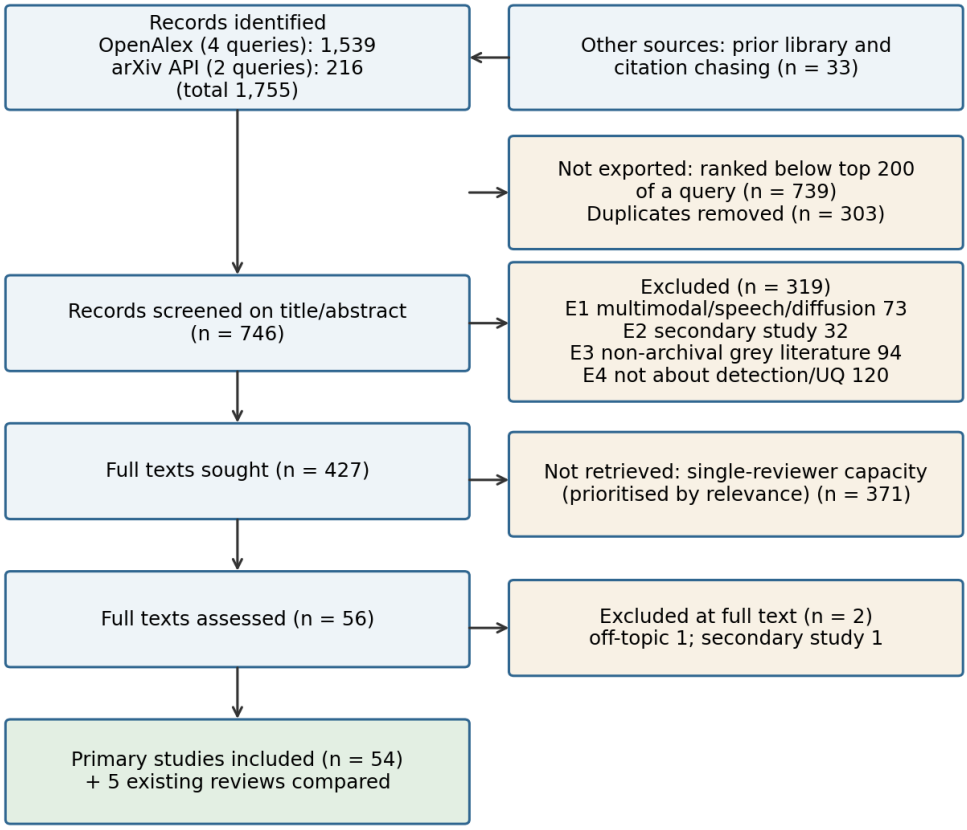


Figure 1. Flow of records through identification, screening and inclusion. All counts come from the PRISMA_Counts sheet of the review workbook.

Figure 1 shows that the largest losses occur before screening (records ranked below the export cut-off) and at full-text retrieval. Both are capacity limits rather than eligibility decisions, so the included set is a prioritised sample of the eligible literature.

4.5 Data extraction and quality appraisal

For each study 31 fields were extracted: bibliographic data, task, method family, method details, datasets, baselines, metrics, main result, evaluation setting, robustness testing, code and data availability, variance reporting, author-stated and reviewer-identified limitations, comparability, evidence strength and RQ mapping. Fields a paper did not report were coded N/R. Every study was read from its full text (abstract, method, experimental setup, result tables and limitations section); appendices were opened only to check a specific item, so the workbook marks every study as read 'Partly'. On 26 October 2026, 19 quality ratings first made from HTML versions were checked again against the PDFs. Every DOI was resolved at

doi.org and its registered title compared with the cited title (all 61 matched). Studies without a publisher DOI are cited with their arXiv DOI.

Quality was judged with the eight questions in Table 3. The answers describe reporting quality. They are not added into a score, and studies are not ranked by them.

Table 4. Quality appraisal of the 54 included studies (number of studies per answer).

Question	Yes	Partial	No	Not reported
Q1 Objective clear	54	0	0	0
Q2 Method described	54	0	0	0
Q3 Dataset described	49	5	0	0
Q4 Baselines fair	42	11	1	0
Q5 Evaluation valid	45	9	0	0
Q6 Variance / statistics	15	3	32	4
Q7 Code or data available	43	2	8	1
Q8 Limitations stated	37	6	10	1

Reporting of objectives, methods and data is good: Q1-Q3 are met by at least 49 of 54 studies. The weak item is Q6: only 15 studies report variance, intervals or significance tests, and 36 report none. 9 studies were rated Partial on Q5, mostly because labels came from the same LLM family that generated the text or because chunk labels were derived from token labels. Q4 is Partial for 11 studies, usually because the newest baselines of the same family were not run.

5. Taxonomy of hallucination detectors

Of the 54 studies, 46 propose or evaluate a detector. 6 are benchmarks or evaluation resources [1, 22, 24, 28, 39, 42] and 2 are analysis or methodology papers [17, 52]. The taxonomy was built from the extraction matrix, not taken from an earlier survey: the method-detail column of each detector was coded on four axes (signal source, supervision, granularity and cost class), and categories were merged until each held at least two detectors. Table 5 gives the rule used to assign each category and the number of detectors in it.

Table 5. Taxonomy axes, assignment rules and number of detectors per category (n = 46).

Axis	Category	Assignment rule	Detectors
Signal source	Attention	Main feature is computed from attention weights or attention-head outputs	13
	Sampled consistency	Score depends on agreement among two or more sampled outputs	12
	Hidden states	Main feature is residual-stream or layer activations	10
	External verifier / LLM judge	A separate model checks the output against a reference or retrieved text	6
	Output probabilities	Score uses only token probabilities of the generating model	3
	Self-report	The model states its own confidence or P(True)	2
Supervision	Supervised	Fitted on labelled hallucinations (including few-shot head selection)	25
	Training-free	No new labels needed	21
Granularity	Response	One score per answer	22
	Sentence	One score per sentence	7
	Span / token / entity	Scores for tokens, spans or named entities	8
	Claim	Scores for atomic claims or claim triplets	9
Cost class	Single pass	Reuses the forward pass that produced the text	22

Axis	Category	Assignment rule	Detectors
	Single pass + auxiliary model	As above plus a small fixed model (for example NLI)	1
	Multi-pass	Needs extra generations or forward passes	18
	External model	Calls a separate verifier, encoder or LLM	5

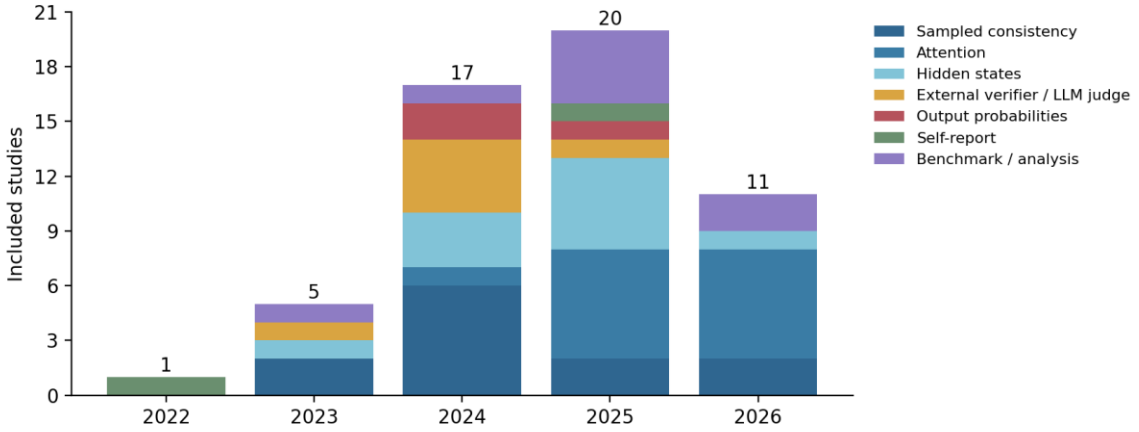


Figure 2. Included studies by publication year and signal source (n = 54).

Figure 2 shows that attention-based detectors are recent: 12 of the 13 appeared in 2025 or 2026, while sampling methods were the main family in 2023 and 2024.

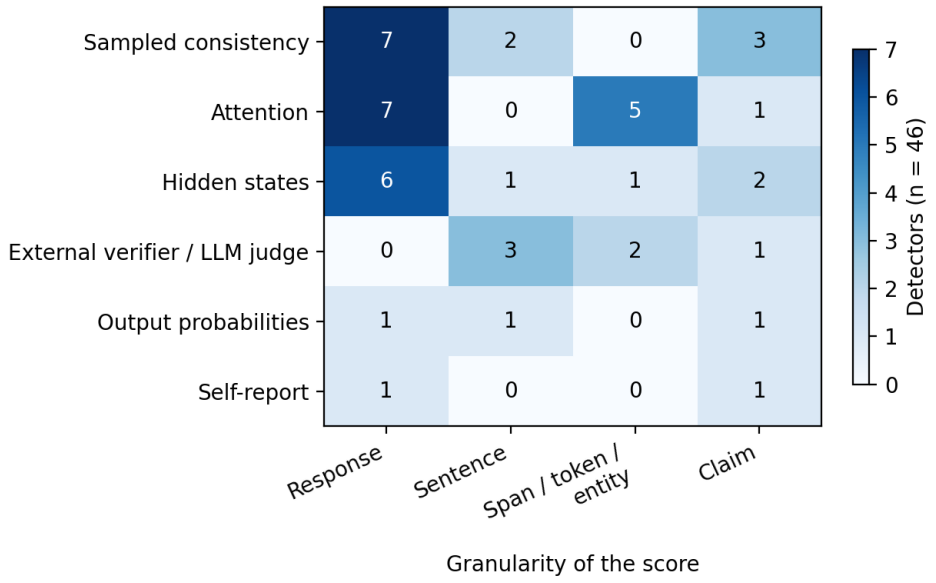


Figure 3. Detectors by signal source and granularity of the score (n = 46).

Figure 3 crosses signal source with granularity. Response-level scoring dominates the three largest families, attention methods are the main group at span level, and the claim column is thin for every family.

5.1 Signal source

Attention (13 detectors [3, 4, 5, 15, 18, 19, 21, 32, 33, 34, 35, 48, 49]) is the largest family in this set. Most attention methods train a probe on attention features such as lookback ratios [15], Laplacian eigenvalues [32], topological divergence [33], sink scores [5] or high-frequency energy [48]. Sampled consistency (12 [6, 7, 10, 11, 12, 23, 25, 26, 29, 46, 51, 54]) is the oldest family and is mostly training-free. Hidden-state methods (10 [13, 14, 16, 20, 27, 36, 37, 43, 45, 53]) train probes on activations. Recent work moves from single layer-token pairs to full activation tensors [37] or residual-stream dynamics [36]. External verifiers (6 [30, 38, 40, 41, 44, 47]) are used mainly for RAG and fine-grained checking. Output-probability methods (3) and self-report methods (2) are small groups.

5.2 Supervision

Supervised detectors (25) slightly outnumber training-free ones (21). Among attention methods, 11 of 13 need labelled data, at least to choose heads [33]. Training-free methods are mostly sampling-based. Three studies reduce the need for labels without removing it: pseudo-labels from uncertainty [53], a probe set of about 100 examples [33] and multi-LLM training with adapters for new models [37].

5.3 Granularity

Response-level scoring dominates (22 of 46). 9 detectors score claims [2, 4, 25, 26, 27, 31, 41, 43, 51], 8 score spans, tokens or entities and 7 score sentences. Claim-level detectors are sampling-based [25, 26, 51], supervised on hidden states or attention [4, 27, 43], an external checker [41], calibrated self-report [31], or probability-based with an auxiliary NLI model [2].

5.4 Cost class

Single-pass detectors (22, plus 1 with an auxiliary model) reuse the forward pass that produced the text. Multi-pass detectors (18) need extra generations or passes (semantic entropy, for example, uses about ten samples [54]). 5 detectors call an external model. Cost class is not the same as measured cost, which Section 7.3 discusses.

5.5 Least-studied combinations and limits of the taxonomy

Crossing the axes shows where evidence is thin. Only 1 detector [2] is training-free, single-pass and claim-level, and it relies on an auxiliary NLI model. Only one attention detector scores claims [4], and it is supervised. No external verifier scores whole responses in this set, and no sampling method scores spans. The taxonomy has two limits. First, the codes are the reviewer's reading of each method description, and hybrid methods (for example attention-guided resampling [35] or attention plus feed-forward signals [18]) were placed by their main signal. Second, the axes describe design, not performance: a category with many detectors is not shown to work better.

6. Comparative analysis

Table A1 (Appendix A) lists all 54 studies with author, year, problem or domain, method, dataset, baseline, metrics, main result and limitations. Results are copied as reported by the authors. Each study was coded for comparability: 7 use a shared protocol that allows direct comparison with other studies (for example the RAGTruth split [24, 40], the LM-Polygraph protocol [28, 43] or MUCH [1]), 42 are

comparable only under conditions (same dataset but different models, labels or splits) and 5 are not comparable. So families are compared by design trade-offs and within-paper results, and numbers from different datasets, settings or hardware are never ranked against each other.

6.1 Single-pass internal signals versus sampling

In several within-paper comparisons, cheap internal-signal probes match or beat sampling methods on the data they were trained on. Semantic entropy probes recover much of semantic entropy's performance from one forward pass [16]. TOHA reports results competitive with SelfCheckGPT and semantic entropy at much lower cost [33]. LettuceDetect, a 150-400M parameter encoder, reports 79.22% example-level F1 on RAGTruth at 30 to 60 examples per second on one GPU [40]. The price is supervision: these detectors need labels for each model, and several authors state that heads or probes do not transfer across architectures [32, 33].

6.2 Attention-based detectors

Attention methods share one idea: a model that is not reading its context or its earlier answer is more likely to hallucinate [15, 17, 18]. They differ in which statistic they read, and the reported gains between them are small. Binkowski et al. [5] report sink-score probes best in 23 of 28 model-dataset pairs, but most gains over Lookback Lens and spectral features are within one standard deviation. Only one attention detector is training-free and single-pass [3], and it scores responses. Only one attention detector scores claims [4], and it is supervised.

6.3 Claim-level and fine-grained detection

Fine-grained evaluation changes conclusions. RefChecker reports that checking claim triplets beats response-level checking by about 10 macro-F1 points on the same responses [41]. Zhang et al. find that models that look well calibrated per response are poorly calibrated per atomic claim [31]. HalluEntity shows that token-probability scores over-predict hallucinations at entity level (AUROC about 0.54-0.62) while context-aware scores reach about 0.77-0.78 across 17 LLMs [42]. MUCH [1] and LM-Polygraph [28] are the only harnesses in this set that evaluate claim-level detectors in more than one language under one protocol.

6.4 External verifiers

Verifier pipelines (FActScore [22], FAVA [30], RefChecker [41], ANAH-v2 [38]) can say which claim is wrong and why, and they work with closed models. They depend on a reference source and on a strong LLM. ANAH-v2 reports that a 7B annotator exceeds GPT-4 on its own test set (F1 89.30 vs 87.11), but the gold labels began as GPT-4 pre-annotations, which the authors note can bias similarity metrics [38].

7. Datasets, metrics and evaluation analysis

7.1 Datasets

Short-answer QA sets dominate. TriviaQA is used by 16 studies, followed by biography sets built on FActScore, WikiBio or similar entity lists (12), SQuAD (8), Natural Questions (8), maths reasoning (8),

CoQA (7) and RAGTruth (7) (Figure 4). Long-form factuality sets such as LongFact appear in only 2 studies. Biographies are the main long-form test bed, so long-form evidence is concentrated on one genre.

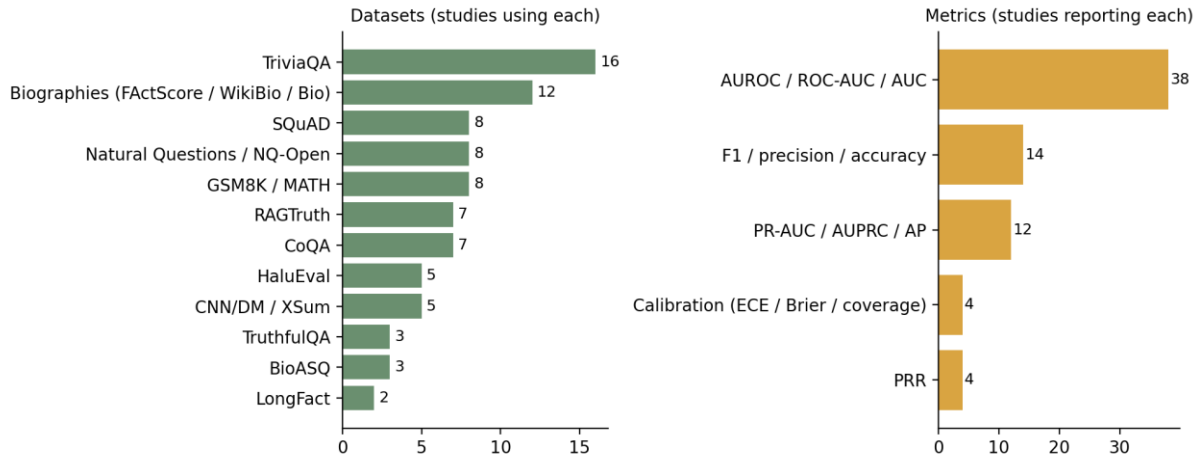


Figure 4. Datasets and metrics used by the included studies (a study can use several). Counts come from keyword rules over the dataset and metric columns of the extraction matrix.

7.2 Labels and metrics

AUROC is the default metric (38 of 54 studies). F1, precision or accuracy is reported by 14, PR-AUC or average precision by 12, PRR by 4 and calibration measures by 4 (Figure 4). Labels come from very different sources: substring match with a reference answer [45], ROUGE thresholds [6], human annotation [22, 24, 30] and, in 12 studies, GPT-4-class judges or automatic pipelines [1, 2, 4, 5, 15, 20, 28, 31, 32, 34, 38, 39]. At least three studies check judge labels against humans or a second judge [1, 15, 32]. Sanity tests by Zollicoffer et al. show that reasoning-trace detectors can score well by reading answer-level artefacts [52], so label and test design affect the measured skill of a detector.

7.3 Evaluation practice

Figure 5 summarises how the 54 studies report evidence. Code is linked by 41 of 54 (75.9%), but own data is released by only 17 of 54 (31.5%). Variance, intervals or significance tests appear in 15 of 54 (27.8%). Transfer to data, models, tasks or languages other than those used for fitting is tested by 16 of 54 (29.6%); counting one further study that tests controlled perturbations [52], 17 of 54 (31.5%) report some shift test. Compute cost (runtime, throughput, FLOPs or overhead against generation) is measured by 10 of 54 (18.5%), although 18 of the 46 detectors need extra generations. Only 6 of 54 (11.1%) evaluate beyond English, and 7 of 54 (13.0%) are directly comparable with other studies.

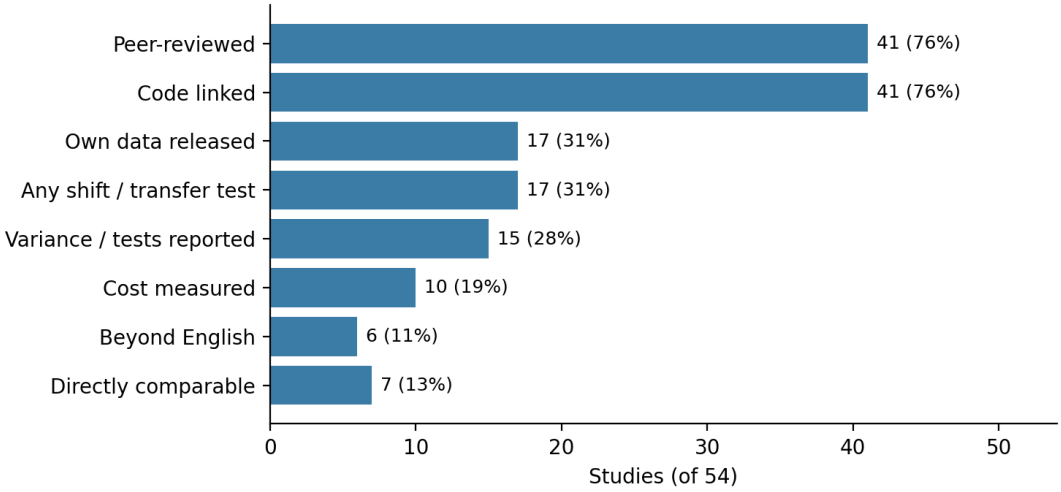


Figure 5. Evaluation and reproducibility indicators across the 54 included studies.

Figure 5 splits into two groups. Things that cost nothing extra to report, such as a code link or a peer-reviewed venue, are common. Things that need extra experiments (repeated runs, transfer tests, timing, other languages) are rare.

8. Challenges and limitations of current methods

Each challenge below is backed by counts from the extraction matrix and by two or three example studies.

- **Access to model internals.** 26 of 46 detectors read attention, hidden states or token probabilities and so need open weights. Attention methods also need eager attention, which is slower than fused kernels [5]. LapEigvals works only on open LLMs [32] and semantic density needs token probabilities that some APIs hide [11]. Closed models allow only sampling or external verifiers.
- **Labels.** 25 of 46 detectors are supervised and need labelled hallucinations for each model and task, and 12 of 54 studies take labels from LLM judges or automatic pipelines. Detectors may then learn the judge's errors: LapEigvals uses gpt-4o-mini labels [32], PsiloQA uses GPT-4o for spans [39] and ANAH-v2 starts from GPT-4 pre-annotations [38].
- **Compute cost.** 18 detectors need several passes and 5 call an external model, but only 10 of 54 studies measure cost. Semantic entropy uses about ten samples [54], conformal abstention needs O(k^2) extra inferences [29] and Finch-Zk samples several different models [46].
- **Granularity mismatch.** 22 of 46 detectors score whole responses, yet long answers mix true and false claims. Claim triplets change results by about 10 macro-F1 points [41], atomic-level calibration is worse than response-level calibration [31], and claim segmentation adds its own error [1].
- **Comparability.** 42 studies are comparable only under conditions and 5 not at all. A gain of 0.01-0.03 AUROC within one standard deviation [5], or RAGTruth results on different base models [18, 33, 40], cannot be ranked across papers.
- **Generalisation.** Only 16 of 54 test transfer. Probes trained on one model's output lose accuracy on another's (AUROC 0.726 in-domain vs 0.625 out of domain for a 3B model [27]). Heads chosen for one architecture do not carry over [32, 33].

9. Research gaps and future directions

Table 6 lists the gaps. Each is supported by counts from the extraction matrix and comes with a proposed study. Strong means the pattern holds across many studies; Moderate means it depends partly on the reviewer's coding.

Table 6. Evidence-based research gaps (counts from the review workbook).

ID	Gap	Evidence (n of N, study IDs)	Why it matters	Proposed study	Strength
G1	Claim-level detection is rare among detectors	9 of 46 (19.6%) method papers score at claim/atomic-fact level (S02, S04, S25, S26, S27, S31, S41, S43, S51); 22 score whole responses, 7 sentences, 8 spans/tokens/entities	One false claim in a long answer is rejected or accepted as a whole; users cannot see what to check	Re-evaluate response-level methods on a claim-level benchmark (MUCH, S01; LM-Polygraph claim tasks, S28) under one protocol	Strong
G2	No training-free, single-pass, attention-based detector works at claim level	Attention-based detectors: 13 of 46 (S03, S04, S05, S15, S18, S19, S21, S32, S33, S34, S35, S48, S49); training-free among them: S03, S35 (S35 needs 3 passes); claim-level: S04 (supervised). Training-free single-pass detectors overall: 4 (S02, S03, S50, S53)	Sampling costs several extra generations; supervised heads/probes need labels per model	Test a label-free attention score at claim level against CCP (S02) and UHeads (S04) on MUCH	Moderate
G3	Compute cost is rarely measured	10 of 54 (18.5%) report runtime, throughput, FLOPs or overhead (S01, S03, S09, S16, S27, S32, S33, S35, S40, S46); 18 of 46 detectors need several generations or passes, 5 need an external model or LLM verifier	Deployment needs overhead relative to generation; MUCH (S01) shows large differences on one GPU	Report overhead as % of generation time per token on stated hardware for every method	Moderate
G4	Transfer to new data, models, tasks or languages is seldom tested	16 of 54 (29.6%) test transfer to data, models, tasks or languages other than those used to fit or tune the method (S04, S08, S13, S15, S16, S20, S27, S32, S33, S34, S36, S37, S38, S39, S43, S54); 1 more (S52) tests controlled perturbations; 17 of 54 (31.5%) report any shift test (Evidence Summary)	Supervised probes are strongest in-domain but need labels for every new setting	Train on one benchmark, test on another with the same protocol; report in-domain vs zero-shot side by side	Strong
G5	Statistical uncertainty of results is seldom reported	15 of 54 (27.8%) report SD, standard error, intervals or significance tests (S02, S05, S10, S11, S13, S16, S20, S21, S27, S28, S29, S32, S33, S49, S54); 3 more report variance only partly (Quality_Check Q6)	Many reported gains are 0.01-0.03 AUROC; without intervals they cannot be interpreted (e.g., S05 gains within one SD)	Bootstrap intervals by generation and paired deltas against each baseline	Strong
G6	Data release lags code release	Code linked: 41 of 54 (75.9%); own data released: 17 of 54 (31.5%) (S01, S04, S06, S07, S13, S20, S24, S25, S26, S30, S38, S39, S41, S42, S47, S48, S54)	Supervised methods cannot be reproduced without their training labels	Release claim-level labels and extracted features with code	Moderate
G7	No shared protocol, so results cannot be ranked across papers	Comparability: Direct 7, Conditional 42, Non-comparable 5 of 54. Metrics: AUROC / ROC-AUC / AUC 38; PR-AUC / AUPRC / AP 12; PRR 4; F1 / precision / accuracy 14; Calibration (ECE / Brier / coverage) 4	Leaderboards built from mixed tables mislead readers	Adopt shared harnesses (MUCH S01, LM-Polygraph S28, RAGTruth S24) and re-run published methods	Strong
G8	Evaluation is almost entirely English	6 of 54 (11.1%) evaluate beyond English (S01, S02, S04, S28, S39, S50)	Deployed LLMs are multilingual; error rates differ by language (S02, S28)	Evaluate on multilingual claim/span sets (MUCH, PsiloQA S39) including non-European languages	Strong
G9	Labels are mostly produced by LLM judges	Studies whose labels come from GPT-4-class judges or automatic pipelines (S01 train, S02, S04, S05, S15, S20, S28, S31, S32, S34, S38, S39): 12 of 54	Detector scores may track the judge's errors, not truth; S52 shows detectors can exploit artefacts	Report judge-human agreement and evaluate on human-labelled test splits	Moderate

Three directions follow from the gaps. First, claim-level evaluation of cheap detectors: G1 and G2 show that the lowest-cost detectors score whole responses, while claim-level detectors are supervised or need sampling. A training-free, single-pass claim-level detector evaluated on MUCH [1] against CCP [2] and UHeads [4] would test whether this trade-off is necessary. Second, shared reporting: overhead as a percentage of generation time, bootstrap intervals with paired differences, and judge-human agreement (G3, G5, G9). Third, transfer and language coverage: train on one benchmark and test zero-shot on another, and include non-English sets such as MUCH and PsiloQA [39] (G4, G8).

10. Limitations of this review

One reviewer screened and extracted all records. Screening decisions were proposed with keyword rules and software assistance and are marked in the workbook for confirmation by the author. No second reviewer checked them. Scopus and Web of Science were not searched. Only the top 200 records per OpenAlex query were exported, and 371 eligible records were not retrieved in full text, so the included set is a prioritised sample of the eligible literature and the counts describe that sample. Appendices were not read in full, so some variance or cost reporting placed only in appendices may be missed, and the counts in Section 7.3 are lower bounds for those items. Taxonomy codes are the reviewer's judgments from method descriptions. 13 of 54 studies are preprints. Finally, the author develops a claim-level attention detector as a separate course project. That work is not among the included studies, and gap G2 was derived from the counts, not from that project.

11. Conclusion

RQ1. Six signal families appear across 46 detectors. Attention (13) and sampled consistency (12) are the largest, followed by hidden states (10). Supervised (25) and training-free (21) detectors are close in number, 22 of 46 score whole responses and 9 score claims.

RQ2. Short-answer QA (TriviaQA in 16 studies) and biographies (12) dominate, AUROC is reported by 38 of 54, and 12 studies take labels from LLM judges or automatic pipelines. Only 7 of 54 studies are directly comparable with others, so results should not be ranked across papers.

RQ3. 18 detectors need several passes and 5 need an external model, but only 10 of 54 (18.5%) studies measure cost.

RQ4. Transfer is tested by 16 of 54 (29.6%) studies and non-English evaluation by 6 of 54 (11.1%). Where transfer is tested, supervised probes lose accuracy outside their training data [27, 43], while multi-LLM training helps [37].

RQ5. Code is often released (41 of 54 (75.9%)), data less often (17 of 54 (31.5%)), and statistical uncertainty is reported by only 15 of 54 (27.8%).

The largest gap is in evaluation. Only 7 of 54 studies can be compared directly and only 15 report uncertainty, so most reported improvements cannot be confirmed across papers. The most useful next step is to re-run existing detectors, including cheap single-pass ones, at claim level on a shared multilingual harness such as MUCH, reporting overhead against generation time and bootstrap intervals for every comparison.

Declarations

Author contributions. Adithyan C S S: conceptualisation, search, screening, data extraction, quality appraisal, analysis, figures and writing.

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Competing interests. The author is developing a claim-level hallucination detector in a separate course project (see Section 10).

Data availability. The review workbook (search log, screening sheet, PRISMA counts, extraction matrix, quality check, evidence summary and gap table, listing every screened record) and the exports of the 26 October 2026 search re-run are supplied as supplementary material.

Use of AI tools. See the separate AI-use declaration submitted with this manuscript.

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Appendix A. Comparison table of the included studies

Results are as reported by the authors under each study's own setting. They are not comparable across rows unless the Comparability column of the workbook says Direct, and rows are ordered by study ID, not by performance. Limitations are author-stated; where a paper states none, the reviewer-identified limitation is given after 'Reviewer:'.

Table A1. Comparison of the 54 included studies.

Author	Year	Problem / domain	Method	Dataset	Baseline	Metrics	Results (as reported)	Limitations
Dentan, J. et al. [1]	2026	Long-form generation: Claim-level UQ benchmark for long-form QA, 4 languages (en/fr/es/de)	Benchmark / resource: Releases 24 logits per token plus a deterministic claim segmenter (much_segmenter, ~0.2% of generation time); evaluates existing white-box UQ baselines	MUCH: 4,873 samples, 4 LLMs (Llama-3.2-3B, Llama-3.1-8B, Ministral-8B, Gemma-3-4B); train split auto-annotated, test split of 200 human-annotated samples	CCP, SAR, Max Likelihood and other token-probability baselines	ROC-AUC, PR-AUC, runtime overhead vs generation time	CCP 0.772 ROC / 0.639 PR at 124% overhead; SAR 0.746 / 0.603 at 22.4% overhead	Only four European languages (author-stated)
Fadeeva, E. et al. [2]	2024	Long-form generation: Claim-level fact-checking via token-level UQ; biography generation	Output probabilities: Claim-Conditioned Probability (CCP): uses an NLI model (DeBERTa-large-MNLI) over top-k alternatives at each token to remove uncertainty about claim choice and surface form	Biography generation, 7 LLMs, 4 languages; automatic claim annotation	Max probability, token entropy, P(True) and other UQ baselines	ROC-AUC, PR-AUC	CCP best for every LLM except PR-AUC on GPT-3.5-turbo (P(True) higher); largest gain Jais-13b: +0.07 ROC, +0.09 PR over closest competitor	NLI model was pre-trained for a different use case; claim segmentation uses OpenAI GPT models; part of the results rest on subjective human evaluation; token rather than phrase units; calibration of CCP not studied
Vazhentsev, A. et al. [3]	2026	Multi-task UQ: Sequence-level hallucination detection, white-box LLMs (QA, summarisation, translation)	Attention: RAUQ: detects 'uncertainty-aware' attention heads that reduce focus on preceding tokens when output is wrong; combines them with token confidence in a recurrence; single forward pass, <1% extra compute	12 datasets across QA, summarisation, translation; 9 LLMs	State-of-the-art UQ baselines	PRR	Consistently outperforms UQ baselines; <1% overhead (author claim)	No dedicated limitation section seen; future work: extend to black-box LLMs, improve calibration, other error types
Shelmanov, A. et al. [4]	2025	Long-form generation: Claim-level hallucination detection in long-form output	Attention: Pre-trained UQ heads: small Transformer modules fed with features from LLM attention maps, trained once per base LLM; released for Mistral, Llama, Gemma 2	Multi-domain training set: 800 texts, 18,852 claims, Mistral-7B-Instruct-v0.2 (as stated in text)	Unsupervised UQ methods (e.g. token probability, CCP-style)	PR-AUC, unsupported claims positive	Mistral-7B-Instruct-v0.2, biographies (in-domain): UHead 0.66 vs CCP 0.50, Factoscope 0.61, Lookback lens 0.56, SAPLMA 0.54. Out-of-domain: UHead best on 6 of 7 domains (landmarks 0.34 vs SAPLMA 0.35). Gemma-2-9B trained on English: Russian 0.58, Chinese 0.54, German 0.44 vs CCP 0.49 / 0.44 / 0.31	Cannot help if the LLM was trained to give misinformation; works best in-domain; LLM biases can transfer; GPT-4o used for claim extraction and labels
Binkowski, J. et al. [5]	2026	Short-form QA: Hallucination detection via attention maps (QA, summarisation, decision support)	Attention: SinkProbe: classifier on attention-sink scores; shows reliance on sinks with large value-vector norms and relates earlier attention methods to sink scores mathematically	7 benchmarks: GSM8K, UMWP, TruthfulQA, TriviaQA, NQ-Open, SQuADv2, HaluEvalQA; 4 LLMs (Llama3.2-3B, Phi3.5-4B, Llama3.1-8B, Mistral-Nemo-12B); correctness labels from an LLM judge with selective manual review	Prior attention-based detectors (e.g. Lookback Lens, topological divergence)	ROC-AUC, mean ± SD over 5-fold cross-validation	Best in 23 of 28 model-dataset pairs (authors). Example Llama3.2-3B: GSM8K 0.845±0.017 vs LookbackLens 0.835±0.017; TruthfulQA 0.785±0.059. Relative gains mostly +0.1% to +4.9%; worse in 5 pairs (e.g. UMWP Llama3.2-3B -6.4%)	Needs attention weights (open-weight models only); efficient deployment needs a specialised kernel; sink scores are correlational, not shown causal
Kuhn, L. et al.	2023	Short-form QA: UQ	Sampled consistency: Semantic	CoQA (dev split, about	Length-normalised	AUROC	Semantic entropy AUROC 0.77	Does not protect against

Author	Year	Problem / domain	Method	Dataset	Baseline	Metrics	Results (as reported)	Limitations
[6]		for free-form natural language generation (QA)	entropy: cluster sampled answers by bidirectional entailment and compute entropy over meanings rather than token sequences; unsupervised, single off-the-shelf model	8,000 questions) and TriviaQA (8,000-question subset of train); OPT 2.7B-30B (headline 30B), 10 sampled answers per question; correctness by Rouge-L > 0.3	entropy, lexical-similarity and other baselines		(CoQA) and 0.83 (TriviaQA) at 30B vs 0.66 / 0.79 for number of distinct answers; beats predictive entropy, length-normalised entropy, lexical similarity and p(True); best temperature 0.5; model accuracy 82.3% CoQA, 50.6% TriviaQA	deception; focused on QA (summarisation needs better paraphrase detection)
Manakul, P. et al. [7]	2023	Long-form generation: Sentence-level hallucination detection, black-box	Sampled consistency: SelfCheckGPT: compare a sentence with sampled responses (BERTScore, QA, n-gram, NLI, prompt variants); zero-resource, black-box	238 GPT-3 passages about WikiBio individuals, sentence-level factuality labels	Grey-box token-probability methods and other black-box baselines	AUC-PR, correlation scores at passage level	Hallucination-class AUC-PR 72.96 (random) up to 83.21; factual-class 27.04 (random) up to 53.97	Passages are mostly Wikipedia-style biographies (author-stated)
Kadavath, S. et al. [8]	2022	Short-form QA: Self-evaluation of correctness: P(True) and P(IK)	Self-report: Models propose answers then estimate P(True); separately trained P(IK) predicts whether the model knows the answer; calibration studies across scale	Multiple-choice, true/false, TriviaQA, GSM8K and Python function-synthesis tasks (8,000 functions, 6,000/2,000 split)	N/A (calibration study)	AUROC, Brier score, calibration	e.g. P(IK) AUROC 0.864 / Brier 0.151 on TriviaQA when trained on TriviaQA	Pretrained models only (RLHF policies may be less calibrated); does not separate truth from what humans say; does not address deception; only five sampling-based datasets for generalisation
Duan, J. et al. [9]	2024	Short-form QA: UQ for free-form QA	Output probabilities: SAR: reweights token and sentence uncertainty by semantic relevance (measured by effect of removing a token on meaning)	Free-form QA: reading comprehension, science and medical QA (SciQ, TriviaQA etc.), up to 33B models	Predictive entropy, length-normalised entropy, semantic entropy	AUROC	SAR beats semantic entropy by 7.1% AUROC on average (Rouge-L 0.5); 2-generation SAR AUROC 0.708/0.685 (SciQ/TriviaQA) beats 5-generation baselines	Extra sentence-similarity computation (author-stated)
Nikitin, A. et al. [10]	2024	Short-form QA: UQ for white- and black-box LLMs (QA)	Sampled consistency: Kernel Language Entropy: semantic-similarity kernels over samples, von Neumann entropy; generalises semantic entropy	BioASQ, NQ, SQuAD, SVAMP, TriviaQA	Semantic entropy, discrete SE, P(True), naive entropy, embedding regression	AUROC, AUARC	Outperforms semantic entropy across models and datasets (tables report mean ± SD)	Needs multiple samples, raising cost (author-stated)
Qiu, X. & Miikkulainen, R. [11]	2024	Short-form QA: Response-wise UQ for free-form QA	Sampled consistency: Semantic density: confidence from a probability density over sampled responses in semantic space (NLI-based distances)	Four free-form QA benchmarks (incl. CoQA), 7 LLMs incl. Llama 3, Mixtral-8x22B	Semantic entropy, P(True), degree, NL, NE, PE	AUROC, AUPR	e.g. CoQA Llama-2-13B AUROC 0.783 vs semantic entropy 0.633 and P(True) 0.594	Needs token output probabilities (unavailable for some proprietary LLMs); most test responses are sentence-level; long responses would need claim decomposition
Vashurin, R. et al. [12]	2025	Multi-task UQ: UQ for QA, summarisation, machine translation	Sampled consistency: CoCoA: Minimum Bayes Risk framework combining model confidence with output consistency; CoCoA-Light variant reduces sampling	QA: TriviaQA, CoQA, MMLU, GSM8k; NMT: WMT14 Fr-En, WMT19 De-En; summarisation: XSUM; subsets and prompts follow S28. Base LLMs: Llama 3.1 8B, Mistral 7B, Falcon 3 7B (Gemma 3 12B in appendix)	SP, PPL, MTE, MCSE, MCNSE, DegMat, EigValLaplacian, Semantic Entropy, SAR, P(True)	PRR up to 50% rejection; AUROC for QA in appendix. Quality measures: accuracy, AlignScore, COMET	Best on every task and model under greedy decoding (mean PRR). Llama8b QA: CoCoA-PPL 0.454 vs SAR 0.414, SP 0.409; NMT: CoCoA-SP 0.519 vs SAR 0.455; SUM: CoCoA-PPL 0.387 vs SP 0.328	Effect varies by model/task/domain; few samples used for consistency; quality metrics may miss nuances of correctness
Azaria, A. & Mitchell, T. [13]	2023	Statement truthfulness: Statement truthfulness from LLM hidden states	Hidden states: SAPLMA: feed-forward classifier on hidden-layer activations, tested at several layers	True-false dataset, 6,084 sentences in 6 topics (cities, inventions, elements, animals, companies, scientific facts)	Sentence probability, few-shot prompting	Accuracy, AUC	71% to 83% accuracy depending on base model; OPT-6.7b AUC about 0.76 for last layer	Evaluates statements rather than long-form generation (author-stated)
Chen, C. et al. [14]	2024	Short-form QA: Hallucination detection from	Hidden states: INSIDE / EigenScore: eigenvalues of the covariance of sampled	CoQA, SQuAD, NQ, TriviaQA	Perplexity, LN-entropy, energy, lexical similarity,	AUROC (AUCs, AUCr), PCC	EigenScore reaches 71.9-79.8 AUCr across OPT-6.7B/LLaMA-7B/13B vs lower values for semantic	Needs internal states, so not for black-box models; does not mitigate (author-stated)

Author	Year	Problem / domain	Method	Dataset	Baseline	Metrics	Results (as reported)	Limitations
		internal states (QA)	responses' embeddings as a consistency measure, plus test-time feature clipping		semantic entropy, SentSAR, SelfCheckGPT		entropy and SelfCheckGPT in the digest table	
Chuang, Y.-S. et al. [15]	2024	RAG / context-grounded: Contextual hallucination detection and mitigation (summarisation, document QA)	Attention: Lookback Lens: per-head ratio of attention on context vs newly generated tokens, linear classifier; classifier-guided decoding	Self-generated: LLaMA-2-7B-Chat greedy answers on 1,000 CNN/DM summaries and 2,655 NQ questions; GPT-4o span-level hallucination labels (97% agreement with humans on a 70-example pilot)	Hidden-state classifier, NLI models	AUROC (predefined spans and sliding window of 8); hallucination rate for guided decoding	Predefined spans: Lookback Lens test 91.2 (QA->Sum) / 88.8 (Sum->QA), transfer 85.3 / 82.0; 28th-layer hidden-state classifier test 91.4 / 83.3, transfer 83.6 / 84.7. Sliding window 8: transfer 66.1 / 66.0 vs 57.7 / 58.8. Guided decoding cuts hallucination 9.6% on XSum	Gains from guided decoding bounded by what the LLM samples among 8 candidates; extra inference for multiple candidates; needs 1k-2k annotated examples
Kossen, J. et al. [16]	2024	Short-form QA: Cheap approximation of semantic entropy	Hidden states: Semantic Entropy Probes: probes on hidden states of one generation trained to predict semantic entropy; no sampling at test time	TriviaQA, NQ Open, SQuAD, BioASQ (+ others)	Accuracy probes, semantic entropy	AUROC	Retains high hallucination-detection performance and generalises better OOD than accuracy probes	Notes that SE needs 5-10 generations
Wu, W. et al. [17]	2025	Mechanistic analysis: Mechanistic study of long-context retrieval and its link to hallucination	Analysis / methodology: Identifies 'retrieval heads' (<5% of heads) via needle-in-a-haystack tests; masking them causes hallucination	Needle-in-a-Haystack, CoT reasoning tasks	Random-head masking	Retrieval score, task accuracy	Pruning retrieval heads breaks retrieval and induces hallucination; pruning random heads does not	No Limitations section; defers in-depth studies (e.g. KV-cache compression) to future work
Sun, Z. et al. [18]	2025	RAG / context-grounded: Hallucination detection in RAG	Attention: ReDeEP: decouples use of external context (copying heads) from parametric knowledge (knowledge FFNs); token- and chunk-level; AARF for mitigation	RAGTruth, Dolly (AC)	16 baselines. Mixed (MPE): SelfCheckGPT, Perplexity, LN-Entropy, Energy, Focus. External-confounded (ECP): Prompt, LMvLM, ChainPoll, RAGAS, Trulens, RefCheck, P(True). Parametric-confounded (PCE): EigenScore, SEP, SAPLMA, ITI	AUC, PCC, Accuracy, Recall, F1	RAGTruth AUC: LLaMA2-7B ReDeEP(token) 0.7325, (chunk) 0.7458 vs best baseline RAGAS 0.7290; LLaMA2-13B chunk 0.8244 vs SEP 0.8089; LLaMA3-8B token 0.7522 vs SAPLMA 0.7092 (chunk 0.7285 lower than token). Dolly(AC) LLaMA2-7B chunk 0.7949 vs best baseline 0.7110	N/R (no limitations section in the full text). Reviewer: Needs retrieved context and internal access. AUC gaps over the best baseline are under 0.02 on RAGTruth for LLaMA2-7B/13B with no seeds reported; chunk-level beats token-level on only 2 of 3 models
Ogasa, Y. & Arase, Y. [19]	2025	RAG / context-grounded: Hallucinated span detection (contextual generation)	Attention: Multi-view attention features (token influence, attention bias, context breadth) fed to Transformer encoder with CRF for sequence labelling	RAGTruth subsets: QA, data-to-text, summarisation	Fine-tuned LLM on the same backbone; Lookback Lens (token-level)	Token-level micro precision, recall, F1	Llama-3-8B-Instruct F1 (QA / Data2Text / Summ.): ours-raw 56.3 / 55.3 / 42.7; ours-norm 55.6 / 55.2 / 44.5; fine-tuning 59.7 / 50.4 / 41.6; Lookback Lens 13.2 / 0.0 / 0.0. Qwen2.5-7B lower	Only two models; English only; needs span-annotated training data
Obeso, O. et al. [20]	2026	Long-form generation: Real-time entity-level hallucination detection in long-form text	Hidden states: Linear and LoRA probes on token activations trained with web-search-grounded entity labels; scaled to 70B	Own long-form annotations; evaluated on LongFact, HealthBench, TriviaQA, MATH	Token perplexity, token entropy, semantic entropy (adapted to long form)	AUC	Llama-3.3-70B: linear probe 0.87, LoRA probe 0.90, semantic entropy 0.71 AUC	Entity-level only; annotation pipeline expensive (author-stated)
van Dijk, G. [21]	2026	Short-form QA: Single-pass hallucination detection from attention (QA)	Attention: KL divergence of each head's attention from uniform, aggregated with a lasso logistic probe	TruthfulQA, TriviaQA, HotpotQA	Existing uncertainty estimation methods	AUROC	TruthfulQA AUROC 0.891-0.906; TriviaQA 0.835-0.846; HotpotQA 0.78-0.80 across Llama-3.2-3B, Qwen3-4B, Mistral-7B	No causal link shown between attention patterns and errors; lasso probe is hard to interpret; needs attention weights, so not usable on black-box models
Min, S. et al.	2023	Long-form	Benchmark / resource:	People biographies; 6,500	Human evaluation	Error rate of	ChatGPT scores 58%; automatic	Biographies and Wikipedia

Author	Year	Problem / domain	Method	Dataset	Baseline	Metrics	Results (as reported)	Limitations
[22]		generation: Evaluating factual precision of long-form text (biographies)	FActScore: split text into atomic facts, report share supported by a knowledge source; automatic estimator uses retrieval + an LM	generations from 13 LMs scored automatically		estimator vs humans, FActScore	estimator error <2%; human evaluation would have cost about $26K	only; not applicable to nuanced, open-ended or debatable facts; measures precision, not recall
Zhang, C. et al. [23]	2024	Long-form generation: UQ for long-text generation	Sampled consistency: LUQ (Atomic / Pair): sentence-level consistency across samples; LUQ-Ensemble picks lowest-uncertainty response among models	FActScore-Bio (500 Wikidata biographies) and FActScore-Dis (new, disease entities); 6 LLMs: GPT-4, GPT-3.5, Gemini 1.0 Pro, Yi-34B-Chat, Tulu-2-70B, Vicuna-33B	Existing short-form UQ methods	Pearson and Spearman correlation between FActScore factuality and uncertainty score (response level)	LUQ Pearson on FActScore-Bio: GPT-4 -0.604, GPT-3.5 -0.713, Gemini 1.0 Pro -0.851, Yi-34B -0.738, Tulu-2-70B -0.776, Vicuna-33B -0.718. Best baseline SelfCheck-NLI is close (e.g. -0.845 on Gemini)	Factuality is the only quality aspect evaluated; ambiguous/unanswerable questions not studied; measures overall model uncertainty, not per-instance
Niu, C. et al. [24]	2024	RAG / context-grounded: Word-level hallucination corpus for RAG	Benchmark / resource: RAGTruth: manually annotated responses from several LLMs on QA, data-to-text, summarisation; evaluates detectors and fine-tunes a small LLM	RAGTruth: nearly 18,000 responses	Prompt-based detection with GPT-4 and other detectors	Precision, recall, F1 at response level and character-level span level	Response-level overall F1: prompt GPT-4-turbo 63.4; SelfCheckGPT (GPT-3.5) 58.8; LMvLM 49.4; fine-tuned Llama-2-13B 78.7. Span-level F1: fine-tuned 52.7 vs GPT-4-turbo prompt 28.3. Choosing between two responses with the detector cut hallucination rate by 21.6% to 63.2% for small models	Generic: benchmark may not cover every practical RAG situation
Bouchard, D. et al. [25]	2026	Long-form generation: Comparative study of fine-grained black-box UQ for long-form output	Sampled consistency: Taxonomy of decomposition, unit-level scoring and aggregation; claim-response entailment, graph-based scorers; uncertainty-aware decoding	FactScore-STEM-Geo (400 questions, new), FactScore-Bio, others; 10 LLM-dataset combinations	Sentence-level SelfCheckGPT-style scorers, LUQ	AUROC, AUPRC	Claim-level AUROC 0.67-0.80; claim-response entailment best in 3 of 10 and within 0.01 elsewhere, top AUPRC in all 10	One temperature, NLI model, embedding model and decomposition/grading LLM; only sentence and claim granularity; five LLMs and two long-form QA datasets may not generalise
Sawczyn, A. et al. [26]	2026	Long-form generation: Fact-level black-box hallucination detection	Sampled consistency: FactSelfCheck: facts as knowledge-graph triples; consistency across samples; fact-level correction	WikiBio (GPT-3), FavaMultiSamples (new)	SelfCheckGPT variants	AUC-PR	WikiBio 92.45 AUC-PR (text, max); FavaMultiSamples 48.52 vs SelfCheckGPT-Prompt 46.91; correction raises factual content by 35.5% vs 10.6%	Notes weaknesses of prior work that tuned on evaluation set
Han, J. et al. [27]	2025	Long-form generation: Hallucination detection in long-form generation	Hidden states: Factuality probes on hidden states for paragraphs and atomic claims, single forward pass	Long-form generations from open models up to 405B	Multi-sample approaches (semantic entropy, SelfCheckGPT-style)	AUROC, FLOPs	In-domain AUROC 0.726 (3B) to 0.758 (405B); OOD (GPT-4o-mini outputs) 0.625 (3B) and up; up to 100x fewer FLOPs than sampling	Depends on an auxiliary model to decompose and refine claims (cost, inconsistency); scores claims independently, ignoring dependencies between claims
Vashurin, R. et al. [28]	2025	Multi-task UQ: Benchmark of UQ for LLMs across 11 tasks	Benchmark / resource: LM-Polygraph benchmark: unified implementation of white- and black-box UQ and confidence normalisation methods	Selective QA: CoQA, TriviaQA, MMLU, GSM8k; selective generation: WMT14 Fr-En, WMT19 De-En, XSum (2,000 instances each; MMLU 100 per subject). Claim-level: biographies of 100 people per setting, 2,499 claims (Mistral-7B EN), 1,100 (Jais EN), 1,031 (Jais AR), 3,104 (Vikhr RU), 2,703 (Yi ZH)	Information-based, density-based, sample-diversity and reflexive methods	PRR (selective tasks); ROC-AUC and PR-AUC with unsupported claims positive (claim-level); MSE for calibration	Claim-level: CCP best in every setting except Chinese (ties P(True)); English Mistral-7B CCP 0.74 ROC / 0.46 PR vs MSP 0.65 / 0.33; Arabic 0.73 / 0.30; Russian 0.67 / 0.80; Chinese 0.61 / 0.25. Selective tasks: CCP, MSP, HUQ-MD, SAR, LexSim best white-box; sample-diversity methods best on average	No Limitations section found; authors discuss limits of ROUGE-L as a quality measure
Abbasi-Yadkori, Y. et	2024	Short-form QA: Abstention for	Sampled consistency: Self-consistency similarity scored by	TriviaQA (short answers) and Temporal Sequences	Log-probability-score baselines	Hallucination (error) rate,	Reliably bounds hallucination rate; less conservative than log-	Needs O(k^2) extra inferences for the score; calibration set

Author	Year	Problem / domain	Method	Dataset	Baseline	Metrics	Results (as reported)	Limitations
al. [29]		generative QA with guarantees	the LLM, conformal prediction to bound hallucination rate; calibrated match threshold	from BIG-bench (long responses)		abstention rate	probability baselines on long responses	must be labelled manually
Mishra, A. et al. [30]	2024	Long-form generation: Fine-grained hallucination detection and editing, information-seeking outputs	External verifier / LLM judge: FAVA: retrieval-augmented LM fine-tuned on synthetic error-inserted data; tags 6 hallucination types per sentence and edits them	FavaBench: about 1,000 human-annotated passages (902 used as test) from ChatGPT, Llama2-Chat 7B/70B	ChatGPT and GPT-4 prompted, retrieval+ChatGPT, FActScore (binary)	Sentence-level per-type F1; binary F1; FActScore after editing	Authors report FAVA beats retrieval-augmented ChatGPT and GPT-4 on fine-grained detection (exact F1 in tables)	N/R (no limitations section in the full text). Reviewer: Synthetic training errors may not match natural hallucinations; GPT-4 baseline without retrieval only (cost)
Zhang, C. et al. [31]	2025	Long-form generation: Calibration of long-form factual generation at atomic-claim level	Self-report: Atomic calibration: decompose responses into atomic facts (GPT-4o), elicit generative and discriminative confidence, two fusion strategies	Bios (500 people), LongFact (1,140 questions), WildHallu (7,917 entities); 7 LLMs (Llama3 8B/70B, Mistral 7B/8x7B, Qwen2 7B/57B/72B)	P(true), self-rating, semantic entropy, EigV (response level)	ECE, Brier, AUROC (atomic); Spearman (macro)	LLMs are worse calibrated at atomic level than at response level; fusion improves calibration	Factuality only; open-source models only
Binkowski, J. et al. [32]	2025	Short-form QA: Hallucination detection in QA from attention maps	Attention: LapEigvals: top-k eigenvalues of the Laplacian of attention maps (as graphs) fed to a probe	7 QA datasets (NQ-Open, TriviaQA, CoQA, SQuADv2, HaluEvalQA, TruthfulQA, GSM8K); labels by gpt-4o-mini judge, agreement checked with gpt-4.1	AttentionScore, attention-feature probes, hidden-state probes, EigenScore, semantic entropy	AUROC (mean ± SD)	State of the art among attention-based methods per authors; ablations on layers, temperature, prompts	Needs labels; incompatible across architectures; minimum length (about 100 tokens); open LLMs only; English only
Bazarova, A. et al. [33]	2026	RAG / context-grounded: Contextual (RAG) hallucination detection in QA and summarisation	Attention: TOHA: topological divergence between prompt and response subgraphs of attention graphs on hallucination-aware heads	RAGTruth (MS MARCO, CNN/DM), CoQA, SQuAD, XSum; 5 LLMs (LLaMA-2-7B/13B, LLaMA-3.1-8B, Mistral-7B, Qwen2.5-7B)	Perplexity, max entropy, ReDeEP, HaloScope, LLM-Check, semantic entropy, EigenScore, SelfCheckGPT	ROC-AUC averaged over 5 splits (± reported); significance tests in appendix	State-of-the-art or competitive with minimal annotated data (probe set of about 100)	Head choice may vary across architectures; text only
Matys, P. et al. [34]	2025	RAG / context-grounded: Online contextual hallucination detection in RAG	Attention: AggTruth: four aggregations of attention mass on passage tokens, per head, over 8-token sliding windows; feature selection of heads; classifier	Own token-labelled RAG responses (labels by GPT-4o) on QA and summarisation; several open LLMs	Lookback Lens (best prior)	AUROC (same-task and cross-task)	Stable same-task and cross-task performance, beats Lookback Lens in several settings	Sliding-window attention models limit context length
Liu, Q. et al. [35]	2025	Short-form QA: Zero-shot hallucination detection in factual QA	Attention: AGSER: split the query into attentive and non-attentive parts by attention contribution, regenerate, and compare consistency scores	Books, Movies, Global Country Information (from InterrogateLLM); Llama2-7B/13B, Llama3-8B, Qwen2.5-14B	SBERT, SelfCheckGPT, INSIDE, InterrogateLLM	AUC	Outperforms the zero-shot baselines on 4 LLMs and 3 datasets with 3 forward passes	Open-source models only; still 3 inference passes
Zhang, Z. et al. [36]	2025	Short-form QA: Hallucination detection from hidden-state dynamics	Hidden states: ICR Score: contribution of attention/FFN modules to the residual-stream update per layer; ICR Probe is an MLP on pooled scores	HaluEval, SQuAD, TriviaQA, HotpotQA; Llama-3-8B-Instruct, Qwen2.5-7B, Gemma-2-9B	PPL, LN-Entropy, LLM-Check, SAPLMA, SEP	AUROC	Better than baselines with fewer parameters; cross-dataset tests	Open models only; detection only; needs more complex tasks
Bar-Shalom, G. et al. [37]	2025	Short-form QA: Hallucination detection from full activation tensors	Hidden states: ACT-ViT: Vision-Transformer-style model over layer x token activation tensors, trainable on several LLMs at once with linear adapters	Datasets and LLMs from Orgad et al. (Mistral-7B, Llama-3-8B and others; QA and other generation tasks)	Probing classifiers (single layer-token), other probes	AUROC	Beats probes; benefits from multi-LLM training; zero-shot to unseen datasets; transfers to new LLMs by adapter fine-tuning	Activation tensors are pooled to reduce size, which may discard informative signal
Gu, Y. et al. [38]	2024	Long-form generation: Scalable hallucination	External verifier / LLM judge: ANAH-v2: EM-style iterative self-training that scales an	ANAH test subset; zero-shot on HaluEval and HalluQA	GPT-4, ANAH-7B/20B	F1, accuracy, ROUGE-L, BERTScore	Stage-3 annotator: F1 89.30, accuracy 89.55 vs GPT-4 87.11 / 86.97 (authors)	Annotator still has a non-negligible error margin; evaluated mainly on own data

Author	Year	Problem / domain	Method	Dataset	Baseline	Metrics	Results (as reported)	Limitations
		annotation for long-form QA	annotation set and retrains a 7B annotator each round					plus HaluEval and HalluQA; one backbone (InternLM2-7B); EM-style training sensitive to initialisation
Rykov, E. et al. [39]	2025	Short-form QA: Multilingual span-level hallucination detection	Benchmark / resource: PsiloQA: automatic pipeline (GPT-4o QA from Wikipedia, answers from 24 LLMs without context, GPT-4o span annotation); evaluates UQ, LLM taggers and encoders	PsiloQA: 14 languages; 63,792 train / 3,355 val / 2,897 test	UQ methods, LLM-based tagging, fine-tuned encoders	Span-level IoU / AP (as in Mu-SHROOM)	Fine-tuned encoders best across languages; cross-lingual transfer works	Single annotator model (GPT-4o) bias; QA task only
Kovacs, A. & Recski, G. [40]	2025	RAG / context-grounded: Token-level hallucination detection for RAG	External verifier / LLM judge: LettuceDetect: ModernBERT (8k context) token classifier over context-question-answer triples	RAGTruth	Luna and other encoders, prompt-based LLM detectors	Example-level and span-level F1	Example-level F1 79.22%, +14.8% over Luna; 30-60 examples/s on one GPU (authors)	N/R (no limitations section; future work: more datasets, languages and architectures). Reviewer: Single benchmark; English only
Hu, X. et al. [41]	2024	Long-form generation: Reference-based fine-grained hallucination checking	External verifier / LLM judge: RefChecker: extract claim-triplets, check each against a reference under zero, noisy and accurate context settings	2.1k responses from 7 LLMs, 11k annotated claim-triplets	RoBERTa-NLI, AlignScore, GPT-4, Claude 2, SelfCheckGPT, FActScore	Macro-F1 over 3 classes; correlation with humans	Triplet granularity beats response level by about 10 macro-F1 points; 6.8-26.1 points over prior methods	Triplets bias toward local context; plain-text general domain only
Yeh, M.-H. et al. [42]	2025	Long-form generation: Entity-level hallucination detection in biographies	Benchmark / resource: HalluEntity: entity-level labels on ChatGPT biographies; evaluates token-probability UQ across 17 LLMs	HalluEntity: 157 biographies (span-level test set)	Likelihood, entropy, CCP, SAR, Focus	AUROC, AUPRC	Context-aware methods better; token-probability methods over-predict; Focus about 0.77-0.78 AUROC vs likelihood about 0.54-0.62	Built on FActScore biographies and inherits their limits (one task; Wikipedia cannot verify every claim); sample-based methods such as semantic entropy cannot be scored per entity
Vazhentsev, A. et al. [43]	2025	Multi-task UQ: Supervised UQ for selective generation and claim-level fact-checking	Hidden states: Token-level Mahalanobis distance from several layers; linear regression on MD features	11 datasets via LM-Polygraph; claim-level fact-checking as in Fadeeva et al.	Information-based and consistency-based UQ	PRR (selective generation); ROC-AUC, PR-AUC (claims)	Substantial gains over existing UQ; generalises out of domain (authors)	Supervised; only 7-9B models tested (compute)
Choi, S. et al. [44]	2023	RAG / context-grounded: Knowledge-grounded generation (dialogue, summarisation)	External verifier / LLM judge: KCTS: knowledge classifier with RIPA token-level hallucination signal guides MCTS decoding of a frozen LM	Wizard of Wikipedia (unseen topics), CNN/DM	FUDGE, NADO, MCTS, SFT and zero-shot LMs, ChatGPT	Token and knowledge overlap, ROUGE-L, BLEU, UniEval, human eval	KCTS improves knowledge grounding over decoding baselines	Extra compute per token (multiple LM and discriminator passes)
Snyder, B. et al. [45]	2024	Short-form QA: Early hallucination detection in open QA	Hidden states: Classifiers on softmax probabilities, Integrated Gradients attributions, self-attention and FC activations; shows signal before the hallucination appears	TriviaQA and other open-domain QA sets; OPT, Falcon, OpenLLaMA	Single-artifact classifiers	AUROC	Classifiers reach high AUROC; tokens before a hallucination already predict it (authors)	Only factual hallucinations; simple setup in which each output contains a single fact
Goel, A. et al. [46]	2025	Long-form generation: Black-box fine-grained detection and mitigation	Sampled consistency: Finch-Zk: compare segments of responses from diverse models to semantically equivalent prompts; targeted correction	FELM (847 questions), GPQA-diamond (198) for mitigation	GPT-4 judge (vanilla, CoT, RAG), SelfCheckGPT	Precision, recall, F1, balanced accuracy (sentence and response); Pearson/Spearman	Sentence-level F1 49.2 vs 46.8 SelfCheckGPT and 35.4 GPT-4 judge (authors' table)	Consistent hallucinations fool consistency; multi-model sampling cost
Zimmerman, I. et al. [47]	2024	RAG / context-grounded: Enterprise RAG	External verifier / LLM judge: RAGHalu: two sequential binary DeBERTa-v3-large NLI	About 50 annotated historical RAG conversations per brand	ChatGPT zero-shot prompts (3-label and binary), fine-tuned	F1 of the UNSUPPORTED class; inference	RAGHalu-2 average UNSUPPORTED F1 0.93 on brand data and best on open-source data;	Only as useful as the knowledge-base articles and retrieval (ill-fitting articles flag

Author	Year	Problem / domain	Method	Dataset	Baseline	Metrics	Results (as reported)	Limitations
		hallucination detection (customer service)	cross-encoders; tier 1 separates verifiable from non-verifiable (small talk) sentences, tier 2 labels verifiable sentences SUPPORTED or UNSUPPORTED against the retrieved knowledge-base articles	from four brands (bank, telecom, credit union, fintech; responses from GPT-3.5-turbo and four open LLMs) plus re-annotated public hallucination datasets; 80/20 stratified splits	Mistral-7B, t5_xxl_true_nli_mixture, RoBERTa-large-MNLI, Vectara model	latency (ms) and cost per inference	fine-tuned models about 10% F1 above zero-shot prompting; estimated $0.000077 per inference vs $0.00037 for GPT-3.5-turbo on one NVIDIA L4	most statements); built for customer-service RAG and underperforms on paraphrase or summarisation data
Qi, S. et al. [48]	2026	RAG / context-grounded: Contextual hallucination detection in RAG at token level	Attention: High-frequency energy of attention signals over generation steps as features for a lightweight detector	RAGTruth, HalluRAG; LLaMA-2-7B/13B-Chat, Mistral-7B-Instruct	SelfCheckGPT, RefChecker, EigenScore, ReDeEP, Lookback Lens, attention variance and entropy	AUROC / F1 (token and response)	Gains over verification, internal-representation and attention baselines (authors)	N/R (no limitations section in the full text). Reviewer: Single family of older LLMs; supervised
More, S. & Pawar, T. S. [49]	2026	Reasoning / maths: Hallucination in mathematical reasoning	Attention: Temporal dispersion of attention entropy across layers as features for a logistic-regression detector	GSM8K, MATH-500; Qwen2.5 1.5B and 3B	Output entropy and probability features	AUC with significance tests	AUC gains up to +0.076 over output-based baselines (authors)	Maths only; Qwen2.5 only
Chen, K. et al. [50]	2025	Long-form generation: Sentence- and passage-level hallucination detection	Output probabilities: Semantic graph over entities and sentences; uncertainty propagation between entities and contradiction-based calibration with neighbour sentences	WikiBio (238 passages, 1,908 sentences); NoteSum (new Chinese, 200 passages, 1,004 sentences)	SelfCheckGPT variants, token-probability methods	AUC-PR (sentence), Pearson/Spearman (passage)	Clear gains over baselines on both datasets (authors)	N/R (no limitations section; NoteSum dataset is private). Reviewer: Small test sets; NoteSum private
Liu, T. & Wu, Z. S. [51]	2024	Long-form generation: Group-conditional UQ for long-form text	Sampled consistency: Claim-level calibration and output-level conformal prediction, extended to multicalibration and multivalid conformal prediction over demographic subgroups	Bio-NQ: 8,541 entities from Natural Questions; FActScore-style claim labels	Self-consistency, P(True), verbalised confidence	Calibration error, coverage	Marginal guarantees break inside subgroups; group-conditional methods fix them (authors)	N/R (no limitations section in the full text). Reviewer: Biographies only
Zollicoffer, G. et al. [52]	2026	Reasoning / maths: Sanity checks for reasoning-trace hallucination detectors	Analysis / methodology: Force and Remove oracle tests; TRACT lexical trajectory scorer	BBH-Tracking, GPQA Diamond, MATH-500, CausalT5K; 5 LLMs	7 black-box UQ scorers (EMR, NCP, BSC, NCS, NSN, RACE, SEU)	AUROC	Many detectors rely on answer-level artefacts; TRACT is robust under the tests	Tests are necessary, not sufficient; canonical-answer tasks only
Srey, P. et al. [53]	2025	Statement truthfulness: Unsupervised hallucination detection	Hidden states: IRIS: prompt LLM to verify a statement, take contextual embedding, train probe on uncertainty-based soft pseudo-labels	True-False dataset (7 topics), HaluEval2 (5 domains), others; Mistral-7B and other models 0.5B-32B	Zero/few-shot prompting, CoT, EigenScore, SAR, CCS, MIND, SAPLMA (supervised), GPT-4o	Accuracy / AUROC	Average 90.38 on True-False vs 52-63 for EigenScore, SAR, CCS, MIND (authors' table)	Single statements, not passages; interpretability
Farquhar, S. et al. [54]	2024	Short-form QA: Detecting confabulations in QA and paragraph generation	Sampled consistency: Semantic entropy over meaning clusters (bidirectional entailment); discrete variant for black-box; paragraph-level via factoid questions	TriviaQA, SQuAD, BioASQ, NQ-Open, SVAMP; biography paragraphs; LLaMA-2, Falcon, Mistral, GPT-4	Naive entropy, P(True), embedding regression	AUROC, AURAC (with error bars)	Semantic entropy beats baselines across datasets and models (authors)	Only confabulations, not systematic errors; needs about 10 samples